Advantages of Generative Diffusion Models: A Statistical Perspective

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(based on the joint work with A. Markovich and K. Yakovlev)

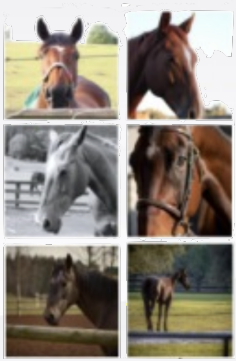
Moscow, October 14, 2026

Generative modelling

Input: i.i.d. random vectors
$Y_1, \ldots, Y_n \in \mathbb{R}^D$

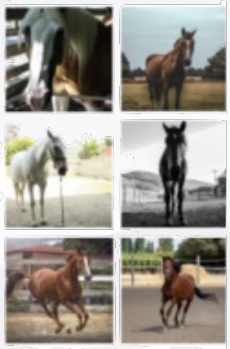
Output: random vector Y
from a similar distribution

training data



Generative model

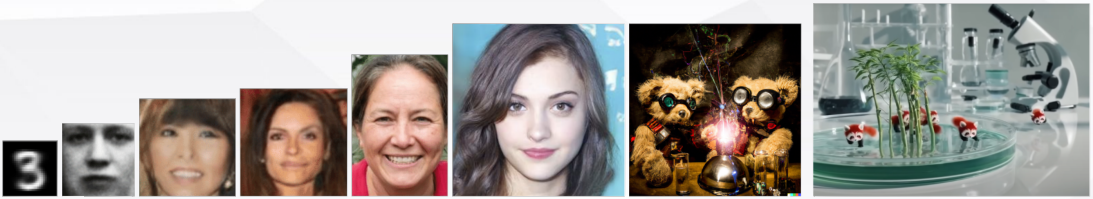
new samples



Generative modelling

Input: i.i.d. random vectors
$Y_1, \ldots, Y_n \in \mathbb{R}^D$

Output: random vector Y
from a similar distribution



2013-2014
Initial attempts to learn generative models; VAE/GAN algorithms

2015-2017
Active development of (primarily GAN-based) algorithms, first convincingly scalable image generation models appear.

2019-2021
Diffusion-based models are introduced and actively developed.

2022
First inspiring large **text-to-image** models appear (based on diffusion models).

2024
Powerful **text-to-video** models appear based on diffusion models.

Figure: Evolution of generative models

Diffusion-based generative models

General idea [Sohl-Dickstein et al., 2015]:

- use a diffusion process to convert any complex distribution into a simple one
- learn a finite-time reversal of this diffusion process



Figure: generative modelling via diffusions. Image courtesy: L. Yang et al. “Diffusion Models: A Comprehensive Survey of Methods and Applications”, ACM Computing Surveys, 2024

Motivation: application to score-based generative models

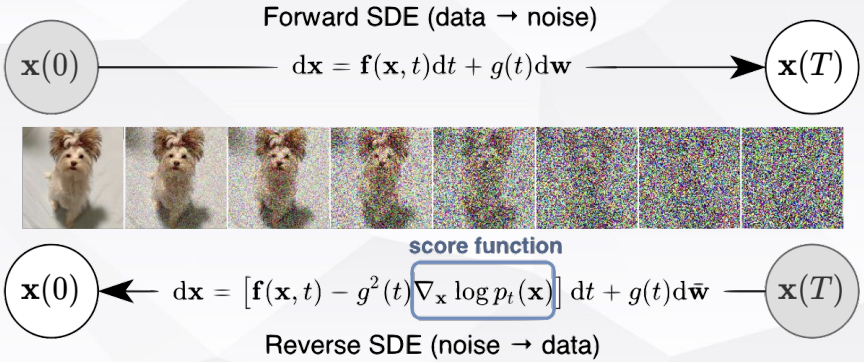


Figure: generative modelling via diffusions. Image courtesy: Y. Song et al. “Score-Based Generative Modeling through Stochastic Differential Equations”, ICLR, 2021

Example: DDPM SDE [Song et al., 2021]

$$\text{forward:} \quad \mathrm{d}X_t = -\frac{\beta_t X_t \,\mathrm{d}t}{2} + \sqrt{\beta_t}\,\mathrm{d}W_t, \quad 0 \leqslant t \leqslant T$$

$$\text{reverse:} \quad \mathrm{d}Z_t = \left(\frac{\beta_t Z_t}{2} + \beta_t \nabla \log \mathrm{p}_{T-t}(Z_t)\right)\mathrm{d}t + \sqrt{\beta_t}\,\mathrm{d}B_t, \quad 0 \leqslant t \leqslant T$$

W_t and B_t are independent Wiener processes

Remark: in what follows, we consider $\beta_t = 2$

Convergence of diffusion models

Fix $0 = t_0 < t_1 < \ldots < t_N \leqslant T$ and consider

$$\mathrm{d}\widehat{Z}_t = \left(\widehat{Z}_t + 2\,\widehat{s}\,(\widehat{Z}_{t_k}, T - t_k)\right)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \quad t_k \leqslant t \leqslant t_{k+1}, 1 \leqslant k \leqslant N$$

Introduce $\gamma_k = t_{k+1} - t_k$ and denote

$$\varepsilon^2_{\text{score}} = \sum_{k=0}^{N-1} \gamma_k\, \mathbb{E}_{X_{t_k} \sim \mathrm{p}_{t_k}} \left\| \nabla \log \mathrm{p}_{t_k}(X_{t_k}) - \widehat{s}\,(X_{t_k}, t_k) \right\|^2$$

Theorem: [Benton et al., 2024]

Assume that $\mathrm{Cov}(X_0) = I_D$ and that $\gamma_k \leqslant \kappa\big(1 \wedge (T - t_{k+1})\big)$ for all $k \in \{0, \ldots, N-1\}$. Then

$$\mathrm{KL}\left(\mathrm{p}_{T-t_N} \,\middle\|\, \mathrm{p}_{\widehat{Z}_{t_N}}\right) \lesssim \varepsilon^2_{\text{score}} + \kappa^2 DN + \kappa DT + De^{-2T}$$

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Question: how to estimate $s^\star(x,t) = \nabla \log \mathrm{p}_t(x)$?

In practice, one fixes a parametric class $\mathcal{S} = \{s_\theta : \theta \in \Theta\}$ (e. g., neural networks) and estimates $s^\star(x,t)$ using i.i.d. samples drawn from p_t

Denoising score matching

OU Forward Process

$$\begin{cases} \mathrm{d}X_t = -X_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t, & 0 \leqslant t \leqslant T \\ X_0 \sim \mathrm{p}_0, & X_0 \in \mathbb{R}^D \end{cases}$$

OU Backward Process

$$\begin{cases} \mathrm{d}Z_t = (Z_t + 2\nabla \log \mathrm{p}_{T-t}(Z_t))\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t, \\ Z_0 \sim \mathrm{p}_T \end{cases}$$

Denoising Score Matching Objective: for a fixed *stopping time* $t_0 > 0$ we aim to minimize

$$\min_{s \in \mathcal{S}} \int_{t_0}^{T} \mathbb{E}_{X_t} \left\| s(X_t, t) - s^\star(X_t, t) \right\|^2 \mathrm{d}t, \qquad s^\star(y, t) = \nabla \log \mathrm{p}_t(y, t).$$

Theorem: [Vincent, 2011]

Let $s^\star(x,t) = \nabla \log p_t(x)$. Then for any $t \in [t_0, T]$ and $s(x,t)$ it holds that

$$\mathbb{E}_{X_t} \|s(X_t,t) - s^\star(X_t,t)\|^2 = \mathbb{E}_{X_0} \mathbb{E}_{X_t} \left[\|s(X_t,t) - \nabla_{X_t} \log \mathrm{p}_t(X_t \mid X_0)\|^2 \mid X_0 \right] + C(t),$$

Denoising score matching loss:

$$\ell(s, X_0) = \int_{t_0}^{T} \mathbb{E}\left[\|s(X_t,t) - \nabla_{X_t} \log \mathrm{p}_t(X_t \mid X_0)\|^2 \mid X_0 \right] \mathrm{d}t, \quad \nabla \log \mathrm{p}_t(X_t|X_0) = -\frac{X_t - m_t X_0}{\sigma_t^2},$$

where $m_t = e^{-t}$ and $\sigma_t^2 = 1 - e^{-2t}$. Given $Y_1, \ldots, Y_n \overset{i.i.d.}{\sim} \mathrm{p}_0$, DSM estimate is

$$\widehat{s} \in \operatorname*{argmin}_{s \in \mathcal{S}} \left\{ \frac{1}{n} \sum_{i=1}^{n} \ell(s, Y_i) \right\}$$

Denoising score matching

Theorem: [Oko et al., 2023]

Assume that p_0 is supported on $[-1,1]^D$ and bounded away from 0 and ∞ on this set. Suppose that p_0 belongs to the Besov space $B^{\beta}_{p,q}([-1,1]^D)$ and is infinitely differentiable inside $[-1,1]^D$. Then the denosing score matching estimate $\widehat{s}$ satisfies

$$\mathbb{E}_{Y_1,\dots,Y_n} \int_{t_0}^{T} \mathbb{E}_{X_t} \left\| \widehat{s}(X_t,t) - s^\star(X_t,t) \right\|^2 \mathrm{d}t = \mathcal{O}\left(\frac{\log^{16}(n)}{n^{2\beta/(2\beta+D)}} \right),$$

provided that the class of feed-forward neural networks is chosen properly

Remark. Kernel-based estimates achieve similar rates of convergence, see [Wibisono et al., 2024], [Zhang et al., 2024]

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Question. Can we escape the curse of dimensionality?

Example: MNIST handwritten digits data set

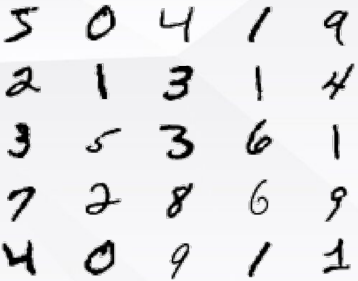


Figure: Handwritten digits (28x28 pixels)

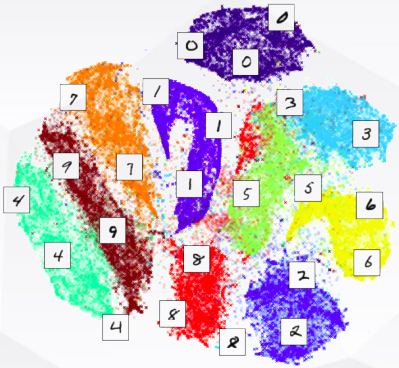


Figure: Embedding into $\mathbb{R}^2$

High-dimensional observations may have a hidden low-dimensional structure!

Assumption

Given a *generator* from the Hölder ball $g^\star \in \mathcal{H}^\beta([0,1]^d, \mathbb{R}^D, H)$, $\|g^\star\|_{L^\infty([0,1]^d)} \leqslant 1$, and $\sigma_{\text{data}} \in [0,1)$, the observed samples $Y_1, \dots, Y_n$ are i.i.d. copies of a random element $X_0 \in \mathbb{R}^D$ generated from the model

$$X_0 = g^\star(U) + \sigma_{\text{data}}\, \xi,$$

where $U \sim \text{Un}([0,1]^d)$ and $\xi \sim \mathcal{N}(0, I_D)$ are independent

Under the assumption, the density along the forward process for any $t > 0$ is

$$\mathrm{p}_t(y) = (\sqrt{2\pi}\tilde{\sigma}_t)^{-D} \int\limits_{[0,1]^d} \exp\left\{-\frac{\|y - m_t g^\star(u)\|^2}{2\tilde{\sigma}_t^2}\right\} \mathrm{d}u, \quad \tilde{\sigma}_t^2 = m_t^2 \sigma_{\text{data}}^2 + \sigma_t^2$$

Proposition: representation of the true score function

Under Assumption 1 it holds that

$$s^\star(y,t) = -\frac{y}{\tilde\sigma_t^2} + \frac{m_t}{\tilde\sigma_t^2}\left(\frac{\int\limits_{[0,1]^d} g^\star(u)\exp\left(-\frac{\|y-m_t g^\star(u)\|^2}{2\tilde\sigma_t^2}\right)\mathrm{d}u}{\int\limits_{[0,1]^d}\exp\left(-\frac{\|y-m_t g^\star(u)\|^2}{2\tilde\sigma_t^2}\right)\mathrm{d}u}\right), \quad \tilde\sigma_t^2 = m_t^2\sigma_{\text{data}}^2 + \sigma_t^2$$

The class of score estimators is a ReLU-network with a tailored architecture

$$\mathcal{S}(L,W,S,B) = \left\{ s(y,t) := -\frac{y}{m_t^2\sigma^2+\sigma_t^2} + \frac{m_t \mathrm{clip}_2(f(y,t))}{m_t^2\sigma^2+\sigma_t^2} : f \in \mathrm{NN}(L,W,S,B),\ \sigma \in [0,1) \right\},$$

where L is the number of layers, $\|W\|_\infty$ is referred to as a width, S is the number of non-zero parameters, and B is weight magnitude

Theorem: [Yakovlev, P., 2025]

Grant the assumption. Let $T \geqslant 1$ and let the sample size be sufficiently large. Then, for any $\delta \in (0,1)$, with probability at least $(1-\delta)$, the denoising score matching estimate satisfies

$$\int_{t_0}^{T} \mathbb{E}_{X_t} \left\| \widehat{s}(X_t, t) - s^\star(X_t, t) \right\|^2 \mathrm{d}t = \widetilde{\mathcal{O}} \left(\frac{T^2 \operatorname{poly}(D)}{\sigma_{t_0}^2} (n\sigma_{t_0}^2)^{-\frac{2\beta}{2\beta+d}} \log(4/\delta) \right),$$

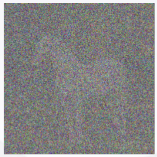
provided that the class of neural networks $\mathcal{S}$ is chosen properly

Inference acceleration: SDE-based vs ODE-based sampling

Low sampling speed!

100s-1000s steps

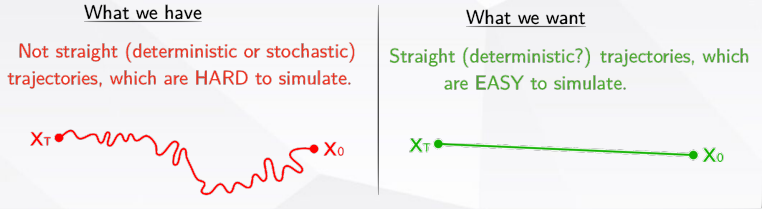


initialize at pure Gaussian

Inference acceleration: SDE-based vs ODE-based sampling

$$\mathrm{d}X_t = f(X_t, t)\mathrm{d}t + g(t)\mathrm{d}W_t, \quad 0 < t < T$$



Probability flow ODE [Song et al., 2021]: remove $\mathrm{d}W_t$ and consider

$$\mathrm{d}X_t = \left(f(X_t, t) - \frac{1}{2}g(t)g(t)^\top \nabla \log \mathrm{p}_t(X_t)\right)\mathrm{d}t, \quad 0 < t < T$$

ODE-based sampling

DDIM sampler (DDPM without stochastic part): $0 = t_0 < t_1 < \ldots < t_K = T$

$$\widehat{X}_T \sim \mathcal{N}(0, I_D), \quad \widehat{X}_{t_{k-1}} = \frac{1}{\sqrt{\alpha_k}} \left(\widehat{X}_{t_k} + \frac{1 - \alpha_k}{2} \widehat{s}(\widehat{X}_{t_k}, t_k) \right)$$

Learning rate schedule:

$$1 - \alpha_1 = \frac{1}{K^{c_0}}, \quad 1 - \alpha_k = \frac{c_1 \log K}{K} \min \left\{ (1 - \alpha_1) \left(1 + \frac{c_1 \log K}{K} \right)^k, 1 \right\}$$

ODE-based sampling

DDIM sampler (DDPM without stochastic part): $0 = t_0 < t_1 < \ldots < t_K = T$

$$\widehat{X}_T \sim \mathcal{N}(0, I_D), \quad \widehat{X}_{t_{k-1}} = \frac{1}{\sqrt{\alpha_k}} \left(\widehat{X}_{t_k} + \frac{1 - \alpha_k}{2} \widehat{s}(\widehat{X}_{t_k}, t_k) \right)$$

Score proximity measures:

$$\varepsilon^2_{\text{score}} = \frac{1}{K} \sum_{k=0}^{K-1} \gamma_k \, \mathbb{E}_{X_{t_k} \sim \mathrm{p}_{t_k}} \left\| \nabla \log \mathrm{p}_{t_k}(X_{t_k}) - \widehat{s}\,(X_{t_k}, t_k) \right\|^2$$

$$\varepsilon_{\text{Jacobi}} = \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}_{X_{t_k} \sim \mathrm{p}_{t_k}} \left\| \nabla^2 \log \mathrm{p}_{t_k}(X_{t_k}) - \nabla \widehat{s}\,(X_{t_k}, t_k) \right\|$$

ODE-based sampling

DDIM sampler (DDPM without stochastic part): $0 = t_0 < t_1 < \ldots < t_K = T$

$$\widehat{X}_T \sim \mathcal{N}(0, I_D), \quad \widehat{X}_{t_{k-1}} = \frac{1}{\sqrt{\alpha_k}} \left(\widehat{X}_{t_k} + \frac{1 - \alpha_k}{2} \widehat{s}(\widehat{X}_{t_k}, t_k) \right)$$

Theorem: [Li et al., 2024]

Assume that p_0 has a bounded support. Then the DDIM sampler with the aforementioned schedule achieves

$$\mathrm{TV}\left(\mathrm{p}_{t_1} \,\|\, \mathrm{p}_{\widehat{X}_{t_1}} \right) \lesssim \frac{D \log^4(K)}{K} + \sqrt{D}\, \varepsilon_{\text{score}} \log^2(K) + D\, \varepsilon_{\text{Jacobi}} \log^2(K)$$

ODE-based sampling: score Jacobian matrix estimation

$$\varepsilon_{\text{Jacobi}} = \frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}_{X_{t_k}\sim \mathrm{p}_{t_k}} \left\| \nabla^2 \log \mathrm{p}_{t_k}(X_{t_k}) - \nabla \widehat{s}\,(X_{t_k}, t_k) \right\|$$

- $\varepsilon_{\text{Jacobi}}$ does not appear in SDE-based sampling
- There are some attempts to estimate the score Jacobian matrix [Genovese et al., 2014], [Sasaki et al., 2018], [Meng et al., 2021]

Question. Can we estimate the score function and its Jacobian matrix? Without the curse of dimensionality?

ODE-based sampling: score Jacobian matrix estimation

Reminder: for any $t > 0$

$$\mathrm{p}_t(y) = (\sqrt{2\pi}\tilde{\sigma}_t)^{-D} \int\limits_{[0,1]^d} \exp\left\{-\frac{\|y - m_t g^\star(u)\|^2}{2\tilde{\sigma}_t^2}\right\} \mathrm{d}u, \quad \tilde{\sigma}_t^2 = m_t^2\sigma_{\text{data}}^2 + \sigma_t^2$$

Lemma: [Yakovlev, Markovich, P., 2025]

Let a function $h : \mathbb{R}^D \to \mathbb{R}^D$ fulfil $|h|_{W^{\alpha,2}(\mathrm{p}_t)} < \infty$ for some integer $\alpha \geqslant 2$. Then it holds that

$$\left\| \|\nabla h\|_F \right\|^2_{L^2(\mathrm{p}_t)} \leqslant \frac{\alpha D^{1-1/\alpha}}{\tilde{\sigma}_t^2} \left(D \|h\|^2_{L^2(\mathrm{p}_t)} + \tilde{\sigma}_t^{2\alpha} |h|^2_{W^{\alpha,2}(\mathrm{p}_t)} \right)^{1/\alpha} \|h\|^{2-2/\alpha}_{L^2(\mathrm{p}_t)}$$

Learning the score is as easy as learning its Jacobian matrix

The class of score estimators. For $\alpha \in \mathbb{N}$ and $t > 0$ define

$$\widetilde{\mathcal{S}}(L, W, S, B) = \Bigg\{ s(x,t) = -\frac{x}{m_t^2\sigma^2 + \sigma_t^2} + \frac{m_t f(x)}{m_t^2\sigma^2 + \sigma_t^2} \ : \ \sigma \in [0,1),\, f \in \mathrm{NN}(L, W, S, B), \\ \max_{1 \leqslant l \leqslant D} |f_l|_{L^\infty(\mathbb{R}^D)} \leqslant C_0,\ \max_{1 \leqslant l \leqslant D} |f_l|_{W^{\alpha,2}(\mathrm{p}_t)} \leqslant C_\alpha \sigma_t^{-2\alpha} \Bigg\}$$

Lemma: [Yakovlev, Markovich, P., 2025]

For any $s \in \widetilde{\mathcal{S}}(L, W, S, B)$, $\alpha \in \mathbb{N}$ such that $\alpha \geqslant 2$, it holds that

$$\left\| \|\nabla s(\cdot, t) - \nabla s^\star(\cdot, t)\|_F \right\|^2_{L^2(\mathrm{p}_t)} \lesssim \frac{D^2 \sigma_t^{-4-4/\alpha}}{(m_t^2 \sigma_{\text{data}}^2 + \sigma_t^2)} \left\| s(\cdot, t) - s^\star(\cdot, t) \right\|^{2-2/\alpha}_{L^2(\mathrm{p}_t)},$$

where the hidden constant is absolute and does not depend on s

Learning the score is as easy as learning its Jacobian matrix

Lemma: [Yakovlev, Markovich, P., 2025]

For any $s \in \widetilde{\mathcal{S}}(L, W, S, B)$, any $\alpha \in \mathbb{N}$, $\alpha \geqslant 2$, and any $t > 0$ it holds that

$$\left\| \|\nabla s(\cdot, t) - \nabla s^\star(\cdot, t)\|_F \right\|^2_{L^2(\mathrm{p}_t)} \lesssim \frac{D^2 \sigma_t^{-4-4/\alpha}}{(m_t^2 \sigma_{\text{data}}^2 + \sigma_t^2)} \left\| s(\cdot, t) - s^\star(\cdot, t) \right\|^{2-2/\alpha}_{L^2(\mathrm{p}_t)},$$

where the hidden constant is absolute and does not depend on s

Corollary. If $\widehat{s}(x, t)$ is sufficiently smooth, then we can estimate $\nabla s^\star(x, t)$ by simple differentiation

Remark. Similarly to score estimations, it is possible to escape the curse of dimensionality, see [Yakovlev et al., 2025] for the details

Further details are in our papers

K. Yakovlev and N. Puchkin

“Generalization error bound for denoising score matching under relaxed manifold assumption”

Proceedings of Machine Learning Research 291:5824-5891, 2025

K. Yakovlev, A. Markovich, and N. Puchkin

“Implicit score matching meets denoising score matching: improved rates of convergence and log-density Hessian estimation”

arXiv:2512.24378

Thank you for attention!

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